QF2104 Tutorial 8

Question 1

Consider $w_{fund2} = \frac{C^{-1}\mu}{1^T C^{-1}\mu}$

i) Show that w_{fund2} is on the minimum variance frontier

Answer: Let us rebuild the minimum variance frontier from the previous tutorial from scratch and derive the two-fund theorem from its principles. Recall

$$\sigma_p^2 = w^T C w$$

Subject to the constraints

$$w^T \mathbf{1} = 1, \quad w^T \mu - \mu_p = 0$$

We then define our Lagrangian

$$L(w_1, w_2, \dots w_n, \lambda_1, \lambda_2) = \frac{1}{2} w^T C w - \lambda_1 \left(w^T \mathbf{1} - 1\right) - \lambda_2 \left(w^T \mu - \mu_p\right)$$

Then we set our partial derivatives against w, λ_1, λ_2 to be 0. Doing so would define λ_1, λ_2 as a simultaneous equation. Where $a = \mathbf{1}^T C^{-1} \mathbf{1}, \quad b = \mu^T C^{-1} \mathbf{1} = \mathbf{1}^T C^{-1} \mu, \quad c = \mu^T C^{-1} \mu$. Given C is a positive definite covariance matrix

$$1 = \lambda_1 a + \lambda_2 b$$

$$\mu_p = \lambda_1 b + \lambda_2 c$$

Thus any w_* that exists on the minimum variance frontier follows:

$$\begin{aligned} w_*(\mu_p) &= \frac{c - b\mu_p}{ac - b^2} C^{-1}\mathbf{1} + \frac{a\mu_p - b}{ac - b^2} C^{-1}\mu \\ &= \frac{ac - ab\mu_p}{ac - b^2} \frac{C^{-1}\mathbf{1}}{1^T C^{-1} 1} + \frac{ab\mu_p - b^2}{ac - b^2} \frac{C^{-1}\mu}{1^T C^{-1}\mu} \\ &= \frac{1}{ac - b^2} \left((ac - ab\mu_p) \frac{C^{-1}\mathbf{1}}{1^T C^{-1} 1} + (ab\mu_p - b^2) \frac{C^{-1}\mu}{1^T C^{-1}\mu} \right) \end{aligned}$$

Now you have expressed the possible portfolio weights in terms of μ_p. Notice if you let $\mu_p = \frac{c}{b}$, the expressions then simplifies to

$$\begin{aligned} w_*(c/b) &= \frac{1}{ac - b^2} \left((ac - ac) \frac{C^{-1}\mathbf{1}}{1^T C^{-1} 1} + (ac - b^2) \frac{C^{-1}\mu}{1^T C^{-1}\mu} \right) \\ &= \frac{C^{-1}\mu}{1^T C^{-1}\mu} = w_{fund2} \end{aligned}$$

Generalized to the two fund portfolio theorem, for any w_* whose portfolio lies on the minimum variance frontier. Where $\alpha + \beta = 1$ and $w_{fund1} = \frac{C^{-1}1}{1^T C^{-1} 1}$, we have

$$w_* = \alpha w_{fund1} + \beta w_{fund2}$$

ii) if $b = \mathbf{1}^T C^{-1}\mu > 0$ show that w_{fund2} is on the efficient frontier

Recall that any portfolio on the **Efficient** frontier must lie on the minimum variance frontier (done in part i) and also have a mean return greater the $\mu_{GMVP} = \frac{b}{a}$. From i. we have $\mu_{fund2} = \frac{c}{b}$

$$\frac{c}{b} - \frac{b}{a} = \frac{ac - b^2}{ab} > 0$$

For any positive definite covariance matrix C, the condition $ac - b^2 > 0$, $a > 0, c > 0$ must be true. (recall our definition of positive definite matrix for a standard nondegenerate case):

$$\forall x \neq \mathbf{0}, \quad x^T C^{-1} x > 0$$

Question 2

In a financial market consisting only of n ($n > 2$) risky assets, g is the global minimum variance portfolio, p is an efficient portfolio, q is the frontier portfolio that is uncorrelated with p, and s is the other frontier portfolio that has the same variance as q. Let the ratio $\frac{\sigma_p^2}{\sigma_g^2} = \lambda$

i) Use the Two-Fund Theorem to show that

$$w_q = \frac{\sigma_p^2}{\sigma_p^2 - \sigma_g^2} w_g - \frac{\sigma_g^2}{\sigma_p^2 - \sigma_g^2} w_p$$

Answer: According to the 2 fund portfolio theorem. Any portfolio w_* can be reconstructed with any 2 other portfolio w_1, w_2 where both portfolios lie on the minimum variance frontier. Hence we have

$$w_q = \alpha w_g + (1 - \alpha) w_p$$

$$\begin{aligned} Cov(r_q, r_p) &= Cov(\alpha r_g + (1-\alpha) r_p, r_p) \\ &= Cov(\alpha r_g, r_p) + Cov((1-\alpha) r_p, r_p) \\ &= \alpha \sigma_g^2 + (1-\alpha)\sigma_p^2 = 0 \end{aligned}$$

Hence we have

$$\alpha = \frac{\sigma_p^2}{\sigma_p^2 - \sigma_g^2}$$

Thus

$$w_q = \frac{\sigma_p^2}{\sigma_p^2 - \sigma_g^2} w_g - \frac{\sigma_g^2}{\sigma_p^2 - \sigma_g^2} w_p$$

ii) Obtain and simplify an expression for μ_q in terms of μ_p, μ_g and λ.

$$\begin{aligned} \mu_q &= w_q r \\ &= \frac{\sigma_p^2}{\sigma_p^2 - \sigma_g^2} w_g r - \frac{\sigma_g^2}{\sigma_p^2 - \sigma_g^2} w_p r \\ &= \frac{\lambda}{\lambda - 1} \mu_g - \frac{1}{\lambda - 1} \mu_p \end{aligned}$$

iii) Show that portfolio q is inefficient.

Just show that the mean returns μ_q is less than μ_g

$$\begin{aligned}\mu_q - \mu_g &= \frac{\lambda}{\lambda-1}\mu_g - \frac{1}{\lambda-1}\mu_p - \mu_g \\ &= \frac{1}{\lambda-1}\mu_g - \frac{1}{\lambda-1}\mu_p \\ &= \frac{1}{\lambda-1}(\mu_g - \mu_p) < 0\end{aligned}$$

To reach our conclusion note 2 core properties. $\lambda > 1$, since σ_g^2 is the minimum variance and variance is positive by definition: our coefficient is positive. Additionally. Since p is and efficient portfolio, assuming p is not exactly portfolio g we have $\mu_p > \mu_g$.

iv) Obtain and simplify an expression for $\frac{\sigma_q^2}{\sigma_g^2}$ in terms of λ only

Recall that

$$\sigma_q^2 = w_q^T C w_q$$

or

$$\sigma_q^2 = \alpha^2\sigma_g^2 + (1-\alpha)^2\sigma_p^2 + 2\alpha(1-\alpha)\sigma_{p,g}$$

Choose the method you prefer

$$\sigma_q^2 = \frac{\lambda^2}{(\lambda-1)^2}\sigma_g^2 + \frac{1}{(\lambda-1)^2}\sigma_p^2 - \frac{2\lambda}{(\lambda-1)^2}\sigma_g^2$$

$$\frac{\sigma_q^2}{\sigma_g^2} = \frac{\lambda^2-2\lambda}{(\lambda-1)^2} + \frac{\lambda}{(\lambda-1)^2} = \frac{\lambda(\lambda-1)}{(\lambda-1)^2} = \frac{\lambda}{(\lambda-1)}$$

v) Obtain and simplify an expression for $\frac{\mu_q-\mu_p}{\mu_s-\mu_p}$ in terms of λ only

Realize that if s has the same variance as q. s belongs to the **Efficient** branch along the minimum variance frontier. Additionally, the minimum variance frontier is symmetrical along $\mu = \mu_g$. Thus,

$$\mu_q = \frac{\lambda}{\lambda-1}\mu_g - \frac{1}{\lambda-1}\mu_p$$

From ii. we have

$$\mu_s = \mu_g + (\mu_g - \mu_q) = 2\mu_g - \mu_q$$

Solving

$$\begin{aligned}\frac{\mu_q-\mu_p}{\mu_s-\mu_p} &= \frac{\mu_q-\mu_p}{2\mu_g-\mu_q-\mu_p} \\ &= \frac{\frac{\lambda}{\lambda-1}\mu_g - \frac{1}{\lambda-1}\mu_p - \mu_p}{2\mu_g - \frac{\lambda}{\lambda-1}\mu_g + \frac{1}{\lambda-1}\mu_p - \mu_p} \\ &= \frac{\frac{\lambda}{\lambda-1}(\mu_g-\mu_p)}{\frac{\lambda-2}{\lambda-1}(\mu_g-\mu_p)} \\ &= \frac{\lambda}{\lambda-2}\end{aligned}$$

Question 3

In a financial market, there are three risky assets whose mean vector and covariance matrix are given by

$$\mu = \begin{pmatrix} 0.4 \\ 0.8 \\ 0.8 \end{pmatrix}, C = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

It is given that $C^{-1}1 = \begin{pmatrix} 0.5 \\ 0 \\ 0.5 \end{pmatrix}$ and $C^{-1}\mu = \begin{pmatrix} 0.1 \\ 0.2 \\ 0.3 \end{pmatrix}$

i) Find the weight vector of the global minimum-variance portfolio.

Answer:

$$w_{GMVP} = \frac{C^{-1}1}{1^T C^{-1}1} = \begin{pmatrix} 0.5 & 0 & 0.5 \end{pmatrix}^T, \quad \mu_{gmvp} = 0.6$$

ii) Find another portfolio of risky assets that lies on the efficient frontier.

$$w_{fund2} = \frac{C^{-1}\mu}{1^T C^{-1}\mu} = \begin{pmatrix} 0.1 & 0.2 & 0.3 \end{pmatrix}^T \div 0.6 = \frac{1}{6}\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}^T, \quad \mu_{fund2} = \frac{11}{15}$$

iii) Use the Two-Fund Theorem to determine whether the following portfolios are efficient portfolios of risky assets.

a) $w = \begin{pmatrix} 5/18 & 1/9 & 1/2 \end{pmatrix}$; b) $w = \begin{pmatrix} 11/18 & -1/9 & 1/2 \end{pmatrix}$; c) $w = \begin{pmatrix} 1/3 & -1/4 & 5/12 \end{pmatrix}$;

2 conditions, all efficient portfolios lay on the minimum variance frontier and must have a return greater than or equal to the mean returns of the GMVP. Notice that we found $\mu_{fund2} = 11/15 > 0.6 = \mu_{gmvp}$. Hence, we simply have to check for 2 conditions for our target w_*. $\exists \alpha \in \mathbb{R} : w_* = \alpha w_{fund2} + (1-\alpha)w_{gmvp}, \quad \alpha \geq 0$ you can simply find α with the following relationships $\mu_* = \alpha\mu_{fund2} + (1-\alpha)\mu_{gmvp}$

- $w_* = \frac{2}{3}w_{fund2} + \frac{1}{3}w_{gmvp}$, since $\alpha \geq 0$, the portfolio is efficient
- $w_* = -\frac{1}{3}w_{fund2} + \frac{4}{3}w_{gmvp}$, since $\alpha < 0$, the portfolio is inefficient even though it exists on the minimum variance frontier
- $\nexists \alpha \in \mathbb{R} : w_* = \alpha w_{fund2} + (1-\alpha)w_{gmvp}$. Hence the portfolio is not on the minimum variance frontier and cannot be efficient.

iv) Find the equation of the efficient frontier for risky assets, giving your answer in the form $\mu = f(\sigma)$

Answer: let $a = \mathbf{1}^T C^{-1}\mathbf{1}, \quad b = \mu^T C^{-1}\mathbf{1} = \mathbf{1}^T C^{-1}\mu, \quad c = \mu^T C^{-1}\mu$. We have from our canonical minimum variance frontier formula:

$$\sigma_p^2 = \frac{a\mu_p^2 - 2b\mu_p + c}{ac - b^2} \quad \mu = \frac{+2b \pm \sqrt{4b^2 - 4a(c - \sigma_p^2(ac - b^2))}}{2a} = \frac{b \pm \sqrt{b^2 - ac + a\sigma_p^2(ac - b^2)}}{a}$$

The efficient frontier corresponds to the positive sqrt of the discriminant

$$\mu = \frac{b + \sqrt{(a\sigma_p^2 - 1)(ac - b^2)}}{a}$$

we should get $\mu = 3/5 + 1/5\sqrt{2\sigma^2 - 2}$

v) Determine the weight vector, mean rate of return and the variance of the rate of return of the tangency portfolio

Answer: This is how I do it, prof may have a slightly different method but the solution remains the same nevertheless

Given a rate of return $r_f = 0.2$ we know that the tangency portfolio occurs in the following condition.

$$max. \frac{\mu - r_f}{\sigma}$$

For business students this expression here is the Sharpe ratio. Our tangency portfolio maximizes the Sharpe ratio, and we can find it multiple ways, with respect to w, μ, σ as long as we have the equation of the minimum variance frontier or the efficient frontier

$$\mu = \frac{b + \sqrt{(a\sigma_p^2 - 1)(ac - b^2)}}{a}, \quad \sigma_p^2 = \frac{a\mu_p^2 - 2b\mu_p + c}{ac - b^2}$$

Hence one possible solution is as follows:

$$S = \frac{\mu_p - r_f}{\sigma_p}$$

Differentiating with respect to σ_p:

$$\frac{dS}{d\sigma_p} = \frac{1}{a}\left[\frac{a(ac - b^2)}{\sqrt{(a\sigma_p^2 - 1)(ac - b^2)}} - \frac{b - ar_f + \sqrt{(a\sigma_p^2 - 1)(ac - b^2)}}{\sigma_p^2}\right] = 0$$

Rearranging:

$$\frac{a(ac - b^2)\sigma_p^2}{\sqrt{(a\sigma_p^2 - 1)(ac - b^2)}} = b - ar_f + \sqrt{(a\sigma_p^2 - 1)(ac - b^2)}$$

$$\frac{ac - b^2}{\sqrt{(a\sigma_p^2 - 1)(ac - b^2)}} = b - ar_f$$

$$(ac - b^2)(a\sigma_p^2 - 1) = \frac{(ac - b^2)^2}{(b - ar_f)^2}$$

Hence, the tangency portfolio variance is

$$\boxed{\sigma_T^2 = \frac{c - 2br_f + ar_f^2}{(b - ar_f)^2}}$$

Now substitute σ_T^2 into the upper branch of the minimum variance frontier:

$$\mu_T = \frac{b + \sqrt{(a\sigma_T^2 - 1)(ac - b^2)}}{a}$$

$$\begin{aligned}
\mu_T &= \frac{b + \frac{ac - b^2}{b - ar_f}}{a} \\
&= \frac{b(b - ar_f) + ac - b^2}{a(b - ar_f)} \\
&= \boxed{\frac{c - br_f}{b - ar_f}}
\end{aligned}$$

By the Two-Fund Theorem we can recover the portfolio weights,

$$w_T = \alpha w_{Fund1} + (1-\alpha) w_{Fund2}$$

where

$$w_{Fund1} = \frac{C^{-1}\mathbf{1}}{a}, \qquad w_{Fund2} = \frac{C^{-1}\mu}{b}$$

Taking expected returns:

$$\mu_T = \alpha \frac{b}{a} + (1-\alpha)\frac{c}{b}$$

$$\alpha = \frac{\mu_T - c/b}{b/a - c/b} = -\frac{ar_f}{b - ar_f}$$

$$1 - \alpha = \frac{b}{b - ar_f}$$

Therefore,

$$\begin{aligned} w_T &= -\frac{ar_f}{b-ar_f}\frac{C^{-1}\mathbf{1}}{a} + \frac{b}{b-ar_f}\frac{C^{-1}\mu}{b} \\ &= \boxed{\frac{C^{-1}(\mu - r_f\mathbf{1})}{b - ar_f}} \end{aligned}$$

If you refer back to prof's notes, you can derive the tangency portfolio through the given formula for w_T the tangency portfolio which follows

$$w_T = \frac{C^{-1}(\mu - r_r 1)}{1^T C^{-1}(\mu - r_r 1)}$$

Which gives us the same final answer. Either ways, for efficiency in test exam its recommended for you to use the given formula to recover your tangency portfolio weights from the given formula above. But if you forget the closed form solution, the longer derivations given above is one way for you to derive the solution from first principles.

Final answer: $w_T = (0, 0.5, 0.5)^T, \mu_T = 0.8, \sigma_T^2 = 1.5$

vi) Find the equation of the Capital Market Line

Answer: The CML is just the line connecting the risk-free asset and the tangency portfolio, extended beyond the tangency portfolio. Hence for the sharpe ratio S

$$\mu = r_f + S\sigma = 0.2 + \frac{0.6}{\sqrt{1.5}}\sigma = \frac{1 + \sigma\sqrt{6}}{5}$$